

C++2: Lesson 4 – Interview

1. Add **summary** at the top and **comments** where necessary.
 - Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

1 **100 points**

5-min challenge: you will get full credit after you reproduce your work during a class or interview.

Create a **Rectangle** class by inheriting from the **Shape** class and write a code to show the area of a rectangle like a sample output:

- a. Implement the **Rectangle** class based on the UML class diagram provided.
- b. Submit a single file containing the test code, **Shape** class, and **Rectangle** class.
- c. Apply your preferred coding style, including indentation and comments, for your final submission.
- d. You may use Ctrl+C/V, (Cmd+C/V) to copy and paste the **Shape** class during your recording or the interview to save time.

Shape
- creator: String
+ Shape(creator: String)
+ getName(): String
+ area(): double



Rectangle
- width: double
- height: double
+ Rectangle(width: double, height: double)
+ area(): double

Note:

***Explain** the code you are writing.

*Perform **two consecutive recordings** for each task.

*Follow the **regular routine** for each interview.

1. **Share the entire screen** (<https://bit.ly/3INpkJV>)
2. **Use only one screen to share (Win + P > PC screen only)**
3. **Display the running programs (Alt + TAB)**
4. Get your **IDE (IntelliJ)** ready

* **Editing** your recordings is considered **cheating** and will not be tolerated.

Please proceed to the next page →

```

class Shape {
protected:
    string creator;
public:
    Shape(string creator) {
        this->creator = creator;
    }
    string getName() {
        return creator;
    }
    virtual double area() {
        cout << "Write code to calculate the area for each shape" << endl;
        return 0.0; // Default implementation
    }
    virtual ~Shape() {}
}

```

Sample
output

Width: ***
Height: ***
Area: ***

File(s) to
submit

C202_1.cpp
C202_1.png,
Clickable YouTube link